Fiscal Year:	FY 2006	**Task Last Updated:**	FY 04/04/2006
PI Name:	Pometto, Anthony Ph.D.		
Project Title:	NASA Food Technology Commercial Space Center		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	apometto@iastate.edu	**Fax:**	FY 515-294-6261
PI Organization Type:	UNIVERSITY	**Phone:**	515-294-9425
Organization Name:	Iowa State University		
PI Address 1:	2312 Food Sciences Building		
PI Address 2:			
PI Web Page:	http://www.ag.iastate.edu/		
City:	Ames	**State:**	IA
Zip Code:	50011-1061	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	07/01/1999	**End Date:**	02/28/2006
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	2
No. of Master's Candidates:	6	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Olson, Dennis		
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The mission of the Food Technology Commercial Space Center (FTCSC) was to lead a national effort to develop foods and food-processing technologies that enhance space missions and advance commercial food products through cooperative efforts with NASA scientists and technologists, commercial companies, and academic researchers. In addition to the $6.6 million in cash and in-kind activities contributed by the center's thirty-four commercial partners, NASA FTCSC research funds were leveraged with affiliate faculty research programs by purchasing equipment that enabled their research program to perform a NASA FTCSC research function. Overall, from 1999-2005, NASA FTCSC-funded equipment, packaging and seed grant requests for center affiliate faculty totaled $30,650, $50,000, and $28,000, respectively. With these equipment purchases, NASA FTCSC gained an additional $4.5 million of non-NASA funds in research efforts in FY 05 giving a six-year total of $14.1 million from a total NASA FTCSC investment of $3,275,000.		

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

NASA FTCSC affiliate faculty and commercial partners worked to develop foods and food-processing technologies that benefit NASA and the public. The Center's commercial partners developed products for space and terrestrial applications. Following is a summary of the commercial partner activities in our final year:
Ajinomoto Co., Inc., after numerous tests and consultations with NASA, terminated the precooked rice product research and turned their attention to developing food products with a calcium absorption promoter. Following positive initial results from the calcium absorption promoter research, the company plans further studies of calcium solubility, intestinal absorption, and potential benefits for bone health. Furthermore, the company funded Paul Flakoll's research project at Iowa State University which examines the metabolic interactions of glutamine supplementation with exercise and insulin resistance in humans.

American Specialty Packaging, Alcoa Flexible Packaging, LLC and KSP (Korean Special Packaging) representatives met in Ames with FTCSC Director Anthony Pometto to discuss NASA's future plans and goals in terms of space food packaging. Since that meeting, the companies have joined FTCSC as a commercial partner and are working on improving space food packaging.

Analytik Jena AG research has resulted in the measurement of changes in antioxidant potential in stored soybeans. Additionally, Analytik Jena has measured the influence of pasteurization and sterilization doses and low pasteurization doses of irradiation on the functional, sensory, and microbial quality of soybeans for soymilk and tofu manufacture. Research also focused on measuring the influence of radiation that soybeans would be exposed to on Mars missions and determining the influence of toasting and storage of full-fat soyflakes on shelf life. In order to conduct this research, the Photochem equipment was used and it was successful in aiding the research.

Arla Foods amba received product approval from NASA for the chocolate milk powder and the thermally-stabilized yogurt and launched the yogurt on Progress 17 in March. Since the launch, Arla has been working to implement a space yogurt production line and to build a special packaging machine.

Biosteam Technologies, Inc. and Selective Micro conducted polymer research and compatibility testing on Selectrocide® for NASA, and completed all EPA testing and approvals for use of the compound as a complete antimicrobial for food, as well as food and non-food surfaces. Also, the company continues to develop a packet towel designed to remove mold and mildew while cleaning surfaces in space vehicles.

Biotecgen s.r.l. continues to work with plant proteinases inhibitors and has determined that they are a significant component of a defense response to plant pathogens and have various technological applications. In addition to that research, the company continues to develop products including liophilised foods for enhanced long-term storage and use in space diets.

California Dried Plum Board researchers are studying the glycemic index values of dried plums, prune juice, and plum juice, and found that dried plum extract can be used as a sweetener in confectionary items. The Board is also sponsoring research that will investigate the effect of dried plums on the acidogenicity of human dental plaque.

Pharmachem Laboratories is performing accelerated stability tests on FruitCare™ chews fortified with ten vitamins. The company is also working to improve their calcium-fortified fruit chews which were found to be less than acceptable during testing at NASA.

The Iowa Soybean Promotion Board and Soyfoods Council sponsored research that was conducted to determine the effects of special soy products, natural antioxidants, wheat proteins, and leavening ingredients to improve the functionality and acceptability of shelf-stable flour tortillas. The research also included evaluating the combined effects of soy products, natural antioxidants, and known anti-staling ingredients in processing, quality, and acceptability of shelf-stable corn tortillas. The ISPB also sponsored Astronaut Clayton Anderson's visit to the Iowa State Fair in 2005.

Kraft Foods Inc. is developing food packaging with reseal technology using advanced materials; light weight packaging of foods; packaging with no loss of strength via nanocomposite technology; and research on nutrient-bone health technology.

MCC, s.a. researched fortification of foods for astronauts during long space flights. The company determined that synthesis of vitamins and minerals will largely depend on the constituents by the food intake. Shelf life considerations impose processing conditions largely incompatible with the preservation of nutritional value as available in the unprocessed form.

Microbial-Vac Systems™, Inc., is working to commercialize their new wet-vacuum, food-prep surface sampling, and separate sample concentration technologies throughout the U.S. and some NATO-member countries. The company also plans to expand research and development efforts for combining sampling and concentration technology with rapid detection systems.

Microcide® has been researching a water-soluble PRO-SAN® sanitizer package and found that the water-soluble pouch product and original powder pouch product resulted in identical microbicidal activities. After additional positive tests, Microcide® is now working with the EPA to obtain approval for the product. For their research in developing food grade sanitizer and disinfecting wipes for long- duration space flights, Microcide® was awarded an SBIR Phase I grant.

Monsanto is working with NASA to research biotech soybeans for their ability to grow under controlled conditions aboard the space station or as appropriate. The company has developed a Space Act Agreement with Kennedy Space Center to provide the Center with unique soybean cultivars for potential benefit to space exploration and in situ food production for habitats on Moon and Mars. Tests will include growing the soybeans in plant growth chambers to assess their potential for space life support applications.

MTI BioTech, Inc., has continued formulation work on products designed to maintain and restore muscle mass as well as improve healing. The company finished testing of primary products containing HMB combined with amino acids and continues its studies involving cancer and trauma patients.

ORBITEC received a SBIR Phase II award in January 2004 for the "Deployable Vegetable Production System (VEGGIE)", a system for growing salad type crops on the International Space Station and other space platforms. The company is currently working with Aubrey Mendonca to develop a HACCP program for VEGGIE.

	Pactech has been focusing on selecting and developing flexible packaging to extend shelf-life from less than 12 months to over two years. The company has set a goal of producing packaging that would extend shelf life to up to 15 years. Another avenue Pactech is researching is the production of Thermal Pouches and will send those to Cape Canaveral for consideration for use in transport of food and pharmaceuticals. ScentSational Technologies has been researching the different applications for their Encapsulated Aroma Release technology and how it could potentially benefit space foods and beverages by making them more aromatic. Current work is focused on scenting the straw spout used for drinking from flexible pouches. Triple “F”, Inc. is focusing their research work on the development and formulation of a textured soy protein-based, meatless Sloppy Joe product, specifically on product preservation through thermal processing. The company was awarded a SBIR in January 2005 to develop a versatile, compact, lightweight, energy efficient, and easily maintained extruder for seed processing in low gravity environments.
Task Progress:	The NASA Food Technology Commercial Space Center (FTCSC) marked its sixth and final year of Center operation. Throughout the year, the Center successfully met all of the requirements for continued funding and continued to make tremendous strides in its mission of developing foods and food-processing technologies that enhance space missions and advance commercial food products. Arla Foods amba received approval for three flavors of its thermally-stabilized yogurt and dehydrated whole milk with probiotic. The thermo-stabilized yogurt was sent to the International Space Station in early 2005 and is currently being purchased and shipped to the soldiers serving in the Middle East. Four companies, ScentSational Technologies, Alcoa Flexible Packaging, American Special Packaging, and Korean Special Packaging, joined FTCSC as new commercial partners. Each of these companies is focusing on space food packaging, which is a number one priority for NASA Johnson Space Center. With this addition, NASA FTCSC currently has 21 active commercial partners who contributed $1,054,337 in in-kind support. A new grant was offered to spur interest and research in the area of packaging. Two packaging projects were funded at $25,000 each in early April. Dr. Lester Wilson was awarded one seed grant and one equipment grant for a total of $16,000 in NASA funding to further his research. Three commercial partners, ORBITEC, Triple “F”, Inc., and Microcide, were awarded SBIR grants from NASA during the past year. FTCSC Director Anthony Pometto presented 16 affiliate faculty and three commercial partner white papers to the NASA Advanced Planning and Integration Office (APIO) Request for Information on November 30, 2004 in Washington, DC. Educational activities, under the direction of Dr. Cheryll Reitmeier, education mission specialist, reached over 2000 Iowa students and teachers with the space food message. Students and teachers learned about food science concepts through space food presentations and workshops led by Reitmeier and Food Science and Human Nutrition Advisor Anne Oldham. Dr. Leonard Williams, Assistant Professor at Alabama A&M University, met with Dr. Aubrey Mendonca at Iowa State University. Mendonca, outreach specialist for FTCSC, hosted Williams and provided lab space for him to learn new food science technologies. Williams is an affiliate faculty member of the Center. In February we learned that NASA Headquarters did not approve the extension of the cooperative agreement. Therefore, this is the final report.
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